

ACM CODE OF ETHICS - Complete Guide with Examples

1. GENERAL ETHICAL PRINCIPLES (7 principles)

Memory aid: "CHABHRF" - Contribute, Harm, Avoid, Be honest, Be fair, Honor, Respect

1.1 Contribute to society and human well-being

- **What it means:** Use your computing skills to benefit people and society
- **Example:** Developing an app that helps elderly people access healthcare services, or creating educational software for underprivileged students
- **Remember:** "Computing for the greater good"

1.2 Avoid harm

- **What it means:** Don't create technology that hurts people physically, mentally, or socially
- **Example:** Refusing to build a surveillance system that violates human rights, or testing software thoroughly to prevent data loss
- **Remember:** "First, do no harm" (like doctors)

1.3 Be honest and trustworthy

- **What it means:** Tell the truth about your work, capabilities, and limitations
- **Example:** Admitting when you don't know something, being truthful about project deadlines, not exaggerating your software's capabilities
- **Remember:** "Truth in tech"

1.4 Be fair and take action not to discriminate

- **What it means:** Treat everyone equally; design inclusive systems
- **Example:** Creating AI algorithms that don't discriminate based on race, gender, or age; ensuring your website is accessible to people with disabilities
- **Remember:** "Tech for all, not some"

1.5 Respect the work required to produce new ideas

- **What it means:** Give credit to others' work; don't plagiarize or steal intellectual property
- **Example:** Citing open-source libraries you use, respecting software licenses, not copying someone else's code without permission
- **Remember:** "Credit where credit's due"

1.6 Respect privacy

- **What it means:** Protect people's personal information and data
- **Example:** Implementing strong encryption, not collecting unnecessary user data, following GDPR regulations
- **Remember:** "Your data is sacred"

1.7 Honor confidentiality

- **What it means:** Keep secrets; don't share confidential information
 - **Example:** Not discussing client projects publicly, protecting trade secrets, respecting NDAs
 - **Remember:** "What's private stays private"
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2. PROFESSIONAL RESPONSIBILITIES (9 principles)

Memory aid: "SMARTER PC" - Strive, Maintain, Authorize, Take review, Evaluate, Respect rules, Perform competently, Public awareness, Computing resources

2.1 Strive to achieve high quality

- **What it means:** Do excellent work in both process and final product
- **Example:** Writing clean, well-documented code; following best practices; thorough testing before deployment
- **Remember:** "Excellence is the standard"

2.2 Maintain high standards of professional competence

- **What it means:** Keep learning and stay ethical in your conduct
- **Example:** Taking courses to learn new technologies, attending workshops, staying updated with industry standards
- **Remember:** "Never stop learning"

2.3 Know and respect existing rules

- **What it means:** Follow laws, regulations, and organizational policies
- **Example:** Complying with copyright laws, following your company's security policies, respecting data protection regulations
- **Remember:** "Rules exist for a reason"

2.4 Accept and provide appropriate professional review

- **What it means:** Welcome feedback and give constructive criticism to others
- **Example:** Participating in code reviews, accepting peer feedback gracefully, reviewing colleagues' work thoughtfully
- **Remember:** "Feedback makes us better"

2.5 Give comprehensive evaluations of computer systems

- **What it means:** Fully assess systems including risks and impacts
- **Example:** Conducting thorough security audits, identifying potential vulnerabilities, assessing environmental impact of data centers
- **Remember:** "See the full picture"

2.6 Perform work only in areas of competence

- **What it means:** Don't take on projects beyond your expertise

- **Example:** Declining a cybersecurity project if you're a web developer without security training; knowing when to consult experts
- **Remember:** "Stay in your lane (or learn first)"

2.7 Foster public awareness and understanding

- **What it means:** Help non-technical people understand technology
- **Example:** Writing clear documentation, explaining technical concepts to clients in simple terms, promoting digital literacy
- **Remember:** "Bridge the knowledge gap"

2.8 Access computing resources only when authorized

- **What it means:** Don't hack or access systems without permission
- **Example:** Not accessing a colleague's account even if you know their password; only using company resources for work purposes
- **Remember:** "Permission required"

2.9 Design and implement secure systems

- **What it means:** Build systems that are both secure and user-friendly
 - **Example:** Implementing two-factor authentication, designing intuitive security features, regular security updates
 - **Remember:** "Security + Usability = Success"
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3. PROFESSIONAL LEADERSHIP PRINCIPLES (7 principles)

Memory aid: "EAMES CR" - Ensure public good, Articulate social, Manage personnel, Articulate policies, Encourage growth, Special care, Recognize infrastructure

3.1 Ensure that the public good is central

- **What it means:** Leaders must prioritize society's benefit
- **Example:** A tech lead rejecting a profitable project that harms user privacy; prioritizing accessibility features
- **Remember:** "Public good > profit"

3.2 Articulate, encourage, and evaluate social responsibilities

- **What it means:** Leaders must communicate and assess ethical obligations
- **Example:** Setting team goals that include social impact metrics, evaluating projects on their societal benefit
- **Remember:** "Lead with values"

3.3 Manage personnel and resources to enhance quality of working life

- **What it means:** Create a good work environment for your team

- **Example:** Preventing burnout, promoting work-life balance, providing growth opportunities, fair compensation
- **Remember:** "Happy team, better work"

3.4 Articulate, apply, and support policies that reflect the Code

- **What it means:** Leaders must create and enforce ethical policies
- **Example:** Establishing a company code of conduct, implementing ethical AI guidelines, creating whistleblower protections
- **Remember:** "Walk the ethical talk"

3.5 Create opportunities for professional growth

- **What it means:** Help team members develop their skills and careers
- **Example:** Providing training budgets, mentorship programs, conference attendance, challenging projects
- **Remember:** "Grow your people"

3.6 Use care when modifying or retiring systems

- **What it means:** Think about consequences when changing or discontinuing systems
- **Example:** Planning proper data migration before shutting down a system, notifying users well in advance, providing alternatives
- **Remember:** "Exit gracefully"

3.7 Recognize and take special care of critical infrastructure systems

- **What it means:** Extra caution with systems society depends on
- **Example:** Being extra careful with healthcare systems, power grid software, financial systems, emergency services
- **Remember:** "Lives may depend on it"

4. COMPLIANCE WITH THE CODE (2 principles)

4.1 Uphold, promote, and respect the principles of the Code

- **What it means:** Follow the Code and encourage others to do so
- **Example:** Speaking up when you see unethical behavior, educating junior developers about ethics, being a role model
- **Remember:** "Be the example"

4.2 Treat violations as inconsistent with ACM membership

- **What it means:** Breaking the Code is serious and unacceptable
 - **Example:** Reporting serious ethical violations, understanding that unethical behavior can end your professional standing
 - **Remember:** "Ethics matter, consequences real"
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SIMPLIFIED MEMORY FRAMEWORK

The 4 Main Sections:

1. **GENERAL** (7) - Personal ethics basics
2. **PROFESSIONAL** (9) - How to work professionally
3. **LEADERSHIP** (7) - How to lead ethically
4. **COMPLIANCE** (2) - Follow and enforce the Code

Quick Numbers to Remember:

- Total: **25 principles**
- Pattern: **7-9-7-2**
- Sections: **4**

The Core Message:

"Use computing skills to benefit society, avoid harm, be honest and fair, respect others, maintain quality, stay competent, and lead responsibly."
